

Remote Stereocontrol with Azaarenes via Enzymatic Hydrogen Atom Transfer

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Strategies for achieving asymmetric catalysis with azaarenes have traditionally fallen short of accomplishing remote stereocontrol, which would significantly enhance accessibility to distinct azaarenes with remote chiral centres. The primary obstacle to achieving superior enantioselectivity for remote stereocontrol has been the inherent rigidity of the azaarene ring structure. Here, we introduce an ene-reductase system capable of modulating the enantioselectivity of remote carbon-centred radicals on azaarenes through a mechanism of chiral hydrogen atom transfer. This photoenzymatic process effectively directs prochiral radical centres located more than six chemical bonds, or over 6 Å, from the nitrogen atom in azaarenes, thereby enabling the production of a broad array of azaarenes possessing a remote γ -stereocenter. Results from our integrated computational and experimental investigations underscore that the hydrogen bonding and steric effects of key amino acid residues are important for achieving such high stereoselectivities.

Introduction

Remote stereocontrol, where a single functional group influences the stereochemical outcome of a reaction at a distant location, represents a formidable challenge in synthetic chemistry due to the decreasing control exerted by the catalyst's chiral environment with increasing distance to the reaction site¹. Yet, the complex chiral structures of enzymes endow them with a natural ability to exercise remarkable control over stereochemical outcomes when forming remote chiral centers²⁻⁴. Azaarenes are nitrogen-containing aromatic heterocycles which are commonly found as functional groups in chiral bioactive molecules such as pharmaceuticals, agrochemicals, and natural products (Fig. 1a)^{5,6}. Despite their utility, the planar rigidity of azaarenes prevents the nitrogen atom from acting as a flexible guide for remote stereocontrol⁷, unlike the more adaptable carbonyl group^{8,9}. As a result, existing methodologies are unable to differentiate the prochiral face distant to a single azaarene group (Fig. 1b)^{7,10,11}.

The advent of photoenzymes has provided an exciting avenue to expand enzymatic catalysis to unexplored reaction domains¹²⁻²⁰. These transformations, especially those involving ketoreductases and ene-reductases, have shown promise in controlling the enantioselectivity of radical prochiral centers in carbonyl group-containing substrates via enantioselective hydrogen atom transfer (HAT).²¹⁻³³ However, as these reductases' native function involves catalyzing the hydrogenation of ketones or activated alkenes containing carbonyl, nitrile, and nitro groups, most new-to-nature reactions rely on the carbonyl group, thus constraining their applicability in asymmetric catalysis³⁴. In this report, we demonstrate the successful utilization of a flavin mononucleotide (FMN)-dependent ene-reductase for enantioselective radical hydroalkylation reactions. The reactions feature α -methyl substituted vinyl arenes and

bromomethyl azaarenes, allowing us to achieve remote stereocontrol with azaarenes. Unlike conventional carbonyl-containing halogenated substrates, bromomethyl azaarenes establish similar hydrogen bond interactions with the ene-reductase. This facilitates effective substrate recognition and activation. Furthermore, the mutual π - π interactions favor charge-transfer (CT) excitation, making the azaarene structure amenable to electron reception from the enzyme, which then generates more stable benzyl radicals. These radicals can be more effectively seized by alkenes, thus aiding the formation of chiral products through an enzyme-catalyzed hydrogen atom transfer mechanism (Fig. 1c). Such chemical processes expand the application range of photoenzymes for remote chiral center synthesis by manipulating the long-range stereochemistry of azaarenes. This approach also enhances the versatility of asymmetric photoenzymatic catalysis concerning substrate functional groups, moving beyond the conventional reliance on carbonyl groups.

Results and discussion

Mechanistically, we propose that the FMN in ene-reductases is reduced by the glucose dehydrogenase (GDH)/nicotinamide adenine dinucleotide phosphate (NADP⁺)/glucose (Glu) system, subsequently forming a CT complex with bromomethyl pyridine. Upon excitation by visible light, the aza-benzylic radical produced via dehalogenation is captured by α -methyl styrene, yielding a prochiral carbon-centered radical. This radical then undergoes enzyme-mediated enantioselective HAT to generate azaarenes bearing a γ -stereocenter (Fig. 1d). After examining various enzymes, visible light wavelengths, additives, and solvents in the reaction (Supplementary Methods and Supplementary Tables 1–6), the old yellow enzyme (OYE1) from baker's yeast emerged as the optimal catalyst, selectively producing the desired product in high yield (80%) and excellent enantioselectivity (97%) (Fig. 1d). This approach is not only applicable to 2-bromomethyl pyridine but also to 4-bromomethyl pyridine (4-BMP) [up to 400 total turnover number (TTN), 95% yield, and 97% enantiomeric excess (ee)] (Extended Data Fig. 1), suggesting that the radical prochiral center, located further from the pyridine's nitrogen atom ($> 6\text{\AA}$), can still be directed for enantioselective hydrogen atom transfer.

Substrate scope for enantioselective hydrogen atom transfer

To demonstrate the synthetic utility of the azaarene-directed remote stereocontrol, we investigated a broad range of azaarenes for hydroalkylation with α -methyl styrene under the optimal reaction conditions (Fig. 2a). The different substituted 2-bromomethyl pyridines underwent the hydroalkylation smoothly to afford the corresponding products (**1** to **14**) with high enantioselectivities (84 to 99% ee), though electron withdrawing substituted 2-bromomethyl pyridines needed more catalyst loading (2 mol%) to give moderate yield (45 to 61%). 4-bromomethyl pyridines were also suitable substrates for the reaction and could give satisfactory outcomes with less catalyst loading (**15**, 0.5 mol%). 4-Bromomethyl pyridines with different substituents generally exhibited high enantioselectivities, and the groups (-Me and -Cl) at the 2-position led to good and moderate yields (77 and 53%, respectively) and excellent enantioselectivities (97 and 93% ee, respectively) (**16** and **17**). The 3-substituted and the disubstituted 4-bromomethyl pyridines gave the desired products (**18** to **20**) with 91 to 99% ee. Other azaarenes including quinoline and pyrimidine also afforded the products (**21** to **23**) with satisfactory enantioselectivities (87 to 90% ee), though the 4-(bromomethyl) quinoline gave a lower yield (**22**, 33%).

Next, we investigated the scope with respect to α -methyl vinyl arenes in the hydroalkylation of 4-bromomethyl pyridines (Fig. 2b). α -Methyl styrene with different substituents afforded the desired products in good to high yields (60 to 96%) with excellent enantioselectivities (90 to 99% ee) (**24** to **32**). Two vinyl arenes (thiophene and benzodioxol) were tolerated, giving high yields (72 and 92%,

respectively) and excellent enantioselectivities (98 and 99% ee, respectively) (**33** and **34**). Furthermore, 2-chloro or 4-chloromethyl pyridines were also suitable substrates for the reaction and afforded the corresponding products (**1** and **15**) in high yields with excellent enantioselectivities (97% ee). It is important to emphasize that substrates containing substituents in the ortho-position of the bromomethyl group may experience steric hindrance, resulting in the generation of methyl azaarene side products due to hydrogen atom transfer to the intermediate aza-benzylic radicals. This can significantly reduce the yield of the desired product.

Mechanistic characterization through mutagenesis

In our quest to shed light on the catalytic mechanism and the structures of the reaction intermediates, we utilized a harmonious blend of experimental and computational methodologies. It is intrinsic to the ene-reductase OYE1 to offer two hydrogen bonding sites for carbonyl-type substrates, namely H191 and N194 (Fig. 3a). Compared to the two lone pairs of electrons on the carbonyl oxygen, the pyridine nitrogen—with its single lone pair—may manifest distinct patterns of hydrogen bonding interactions with residues. To gain deeper insights into the pivotal roles played by residues H191 and N194, we engineered OYE1 H191X and N194X mutants via site-directed mutagenesis and evaluated their catalytic activity and enantioselectivity in hydroalkylation under standardized reaction conditions (Fig. 3b). While mutations at H191 show modest influence on reactivities—with H191A and H191G mutants demonstrating enhanced product yield and high enantioselectivities (99% ee)—mutations at N194 also have a noticeable effect on reactivity. Most N194 mutations lead to a slight, yet noticeable, reduction in both yield and enantioselectivity. These experimental observations suggest that N194 plays a discernible role in substrate interaction and influences reaction activity and selectivity to a certain extent, while the H191 residue appears less involved in forming a hydrogen bond with 4-BMP during the reaction. Further, the lowest energy conformation, ascertained through molecular docking, indicates that 4-BMP is predisposed to interact more with N194, forming an essential hydrogen bond that paves the way for substrate ingress into the reaction site for subsequent transformation (Fig. 3c).

The genesis of chiral HAT was subsequently probed through deuterium labelling experiments. Within the Glu/GDH/NADP⁺ framework, glucose serves as a substrate for GDH, enabling the reduction of NADP⁺ to NADPH. This, in turn, reduces FMN to FMN hydroquinone (FMN_{hq}) in the OYE1 cofactor recycling system (Fig. 3d). The reaction was executed using in situ deuterated FMND, derived from D-glucose-1,2,3,4,5,6-d₇, resulting in a product yield of 39% and a 89% deuterium labelling at the novel stereocenter. When deuterated glucose was amalgamated with an acetate buffer in H₂O, a deuterium labelling ratio of 53% was retained. Conversely, the use of non-deuterated D-glucose in a deuterated buffer led to a deuteration rate (9% D) at the chiral center (see Supplementary Fig. 1 for details). This evidence points towards hydrogen atoms originating primarily from FMN_{hq}, and the in situ generated FMNH(D) potentially exchanging hydrogen with the solution, thereby influencing the deuteration rate. We also took note of the proximal tyrosine residues, namely Y82, Y196, and Y375, due to their close proximity to the reaction site, raising the possibility of their involvement in transferring hydroxyl hydrogen atoms to the substrate and acting as a hydrogen atom shuttle. To clarify their roles, we mutated these residues to phenylalanine, eliminating their hydrogen-donating capabilities, and assessed the resultant mutants under optimal conditions (Fig. 3e). The findings from these experiments, including those from multi-site mutants, revealed that despite modifications, the mutants produced products with comparable enantioselectivities to the wild-type OYE1. Even the Y375F_Y196F_Y82F mutant, while yielding less product (56%), maintained a high enantiomeric

excess (92%). These results led us to conclude that these tyrosine residues do not primarily dictate the chiral HAT process. This supports our hypothesis that the FMN semiquinone³⁵ (FMN_{sq}) is likely responsible for the direct transfer of hydrogen atoms to the substrate, thereby controlling the enantioselectivity of the reaction.

Quantum cluster model computation for mechanistic understanding

To delve into the intricate nuances of the reaction mechanism, we performed a comprehensive computational exploration by a synergistic approach of molecular docking and cluster modeling³⁶⁻³⁸. Initially, potential conformations of 4-BMP in conjunction with the enzyme were scrutinized via AutoDock³⁹, employing the X-ray crystallographic structure of OYE1 as a benchmark. Subsequently, amino acid residues in close proximity to the substrate within the primary shell were isolated for each conformation, and their terminal carbon atoms were fixated (for an exhaustive explanation, see Supplementary Methods). Utilizing the quantum chemical cluster model, the complex with the most favourable energy disposition was ascertained, whereby the extracted 13 amino acid residues played a pivotal role in the construction of an open pocket, functioning in tandem with FMN_{hq} (Extended Data Fig. 2). This structural configuration paved the way for substrate accessibility to the reaction site through hydrogen-bond interactions involving residue N194 (Structures of alternative higher-energy complexes are provided in Supplementary Fig. 3). Following this, studies elucidating the mechanics of radical initiation were initiated. Time-dependent density-functional theory (TDDFT) calculations were conducted to shed light on the fundamental nature of the CT complex between FMN_{hq} and 4-BMP. The calculations underscore that the highest occupied molecular orbital (HOMO) of the CT complex predominantly resides in the π orbital of FMN_{hq} ($\pi^{\text{FMN}_{\text{hq}}}$), while the lowest unoccupied molecular orbital (LUMO) is nearly situated in the π^* orbital of the substrate ($\pi^{4\text{-BMP},*}$) (Fig. 4a)^{40,41}. The combined influence of HOMO and LUMO surpasses 90% in charge-transfer excitation, while the interfragment charge transfer (IFCT) method based on the Multiwfn program⁴¹ revealed that 0.94 electrons are transferred from FMN_{hq} to 4-BMP during absorption. This CT excitation would generate FMN_{sq} directly and cause the substrate's antibonding orbital ($\pi^{4\text{-BMP},*}$) to accumulate electrons, thereby facilitating substrates to undergo debromination to produce aza-benzylic radicals following inter-system crossing (Fig. 4b). The spin density analysis on the **IntA**(triplet) illustrates that the diradical's two electrons are respectively delocalized on the azaaryl ring and the FMN_{sq} plane, which bolsters the ensuing nucleophilic attack and hydrogen atom transfer process (Structures of alternative higher-energy **IntA** are provided in Supplementary Fig. 4).

The genesis of enantioselectivity was subsequently probed computationally with diradical **IntA** serving as the initiating intermediate. Computational insights suggest that styrene in its unbound state tends to interact with the aza-benzylic radical, culminating in **IntB**, which then participates in radical nucleophilic attack via transition states **TSNA**, yielding intermediates **IntC-triplet** (Fig. 4c). As indicated by the deuterium substitution experiment and tyrosine mutation experiment displayed in Fig. 3, the intermediate is poised to receive a hydrogen atom directly from FMN_{sq}, culminating in the generation of the product. This step requires a transition in electron spin state from triplet to singlet. Accordingly, to pinpoint the most stable transition state structures, we executed computations on chiral HAT with singlets, spin-singlets, and triplets, resulting in 29 unique HAT transition state conformations (Supplementary Fig. 6 and Table 7). It was conclusively established that the system exhibits the lowest energy for HAT in the spin-singlet state, while the highest energy for HAT was observed in the triplet state. The spin state transition calculation illustrates the propensity of **IntC-triplet** to give rise to **IntDS-**

triplet and **IntDR-triplet** via swift conformational tautomerism, which subsequently undergoes inter-system crossing to afford precursors **IntDS-spin** and **IntDR-spin** for HAT, respectively. Electron spin density analysis elucidates the specifics of such a spin transition of the intermediate and transition states from triplet to spin singlet (Extended Data Figure 3). The most favourable two spin-singlet transition states, **TSHATS-spin-I** and **TSHATR-spin-I**, for HAT possess larger free energy barriers ($\Delta\Delta G = 21.2$ kcal/mol) relative to other steps, indicating that the HAT process governs both the rate-determining and enantioselectivity-determining steps. Aligning with experimental observations, the computed energy of transition state **TSHATS-spin-I** was 2.7 kcal/mol lower than that of **TSHATR-spin-I** (Fig. 4c), facilitating the favoured (*S*)-product of hydroalkylation (for the determination of absolute configuration, refer to Supplementary Fig. 7).

The computational structures corresponding to the lowest-energy transition states for the primary and secondary enantiomers of the resultant product are detailed in Fig. 4d. These states exhibit substrates that are secured through a hydrogen bond between N194 and azaarene nitrogen, facilitating a swift hydrogen atom transfer from the N-H of FMN_{sq} to the radical prochiral center, which results in product formation. To investigate the cause of significant enantioselectivity, we used the SambVca⁴² online tool to create topographic steric maps, centered on the nitrogen atom of FMN. This approach yields a three-dimensional portrayal of the catalytic pocket, wherein zones accentuated in red depict areas of marked steric hindrance. Our findings revealed that quadrant III, influenced by Y375 and T37, exhibits the greatest steric hindrance, impacting how the substrate approaches the FMN plane during hydrogen bond formation with N194. This interaction guides the phenyl group towards the least hindered quadrant IV, while the methyl group is steered towards the more hindered quadrant III, producing the *S*-configuration product. Further exploration, involving mutagenesis of key residues in the catalytic pocket, suggested a correlation between decreasing steric hindrance at the Y375 site and a decrease in the reaction's enantioselectivity (Extended Data Figure 4, entries 1-6). Interestingly, the T37A mutant showed only a modest decrease in enantioselectivity, unlike other key residues substituted with alanine, which had no noticeable effect (Extended Data Figure 4, entries 7-14). The steric hindrance maps for Y375A and T37A mutants suggest that the mitigation of steric hindrance, particularly at Y375, results in a lesser disparity between quadrants III and IV's steric hindrances. This shift impacts the enzyme's discrimination capability between methyl and phenyl groups, thus driving a reduction in enantioselectivities.

Conclusions

In this study, we have demonstrated a novel approach to achieving remote stereocontrol with a single azaarene group, by utilizing the photoinduced promiscuity of flavin-dependent ene-reductases. Our results highlight the key roles of specific amino acid residues in determining enantioselectivity, emphasizing the unique capabilities of biocatalysts in controlling stereoselectivity at remote sites. We also present a viable strategy for incorporating an azaarene pharmacophore into 'ene'-containing compounds, opening up potential avenues for creating value-added products from biomass-derived alcohols. Importantly, our work addresses a long-standing challenge in asymmetric catalysis by offering a viable strategy for remote stereocontrol with a single functional group. By demonstrating this novel chemistry, we hope to expand the current boundaries of asymmetric catalysis and contribute to the development of more efficient and sustainable synthetic methodologies.

Acknowledgments. This work was funded by the DOE Center for Advanced Bioenergy and Bioproducts Innovation, under the auspices of the U.S. Department of Energy, Office of Science, Office of Biological and Environmental Research (Award Number DE-SC0018420 to H.Z.). The funders had no role in study design, data collection and analysis, decision to publish, or preparation of the manuscript. We acknowledge the Computational Chemistry Commune (<http://bbs.keinsci.com/>) for their support with DFT calculations. Our research benefitted from the computing resources at Delta, the National Center for Supercomputing Applications, enabled by allocation BIO230016 from the Advanced Cyberinfrastructure Coordination Ecosystem: Services & Support (ACCESS) program, funded by National Science Foundation grants #2138259, #2138286, #2138307, #2137603, and #2138296. The Delta research computing initiative, funded by the National Science Foundation (award OCI 2005572), the State of Illinois, and a partnership between the University of Illinois at Urbana-Champaign and its National Center for Supercomputing Applications, is deeply appreciated. We are indebted to Dr. Guangde Jiang and Dr. Haiyang Cui for their invaluable insights. Special thanks to Lucienne T. Burrus and O'Dell Miranda C. for their organizational assistance during the project.

Author contributions: The project was coordinated by H.Z., while H.Z. and M.L.L. jointly conceptualized the project and designed the research experiments. M.L.L. executed the experiments and computational studies. W.H. and Z.Z. were responsible for synthesizing some substrates, and Y.Y. contributed to the construction of the mutants.

Competing interests: The authors declare no competing interests.

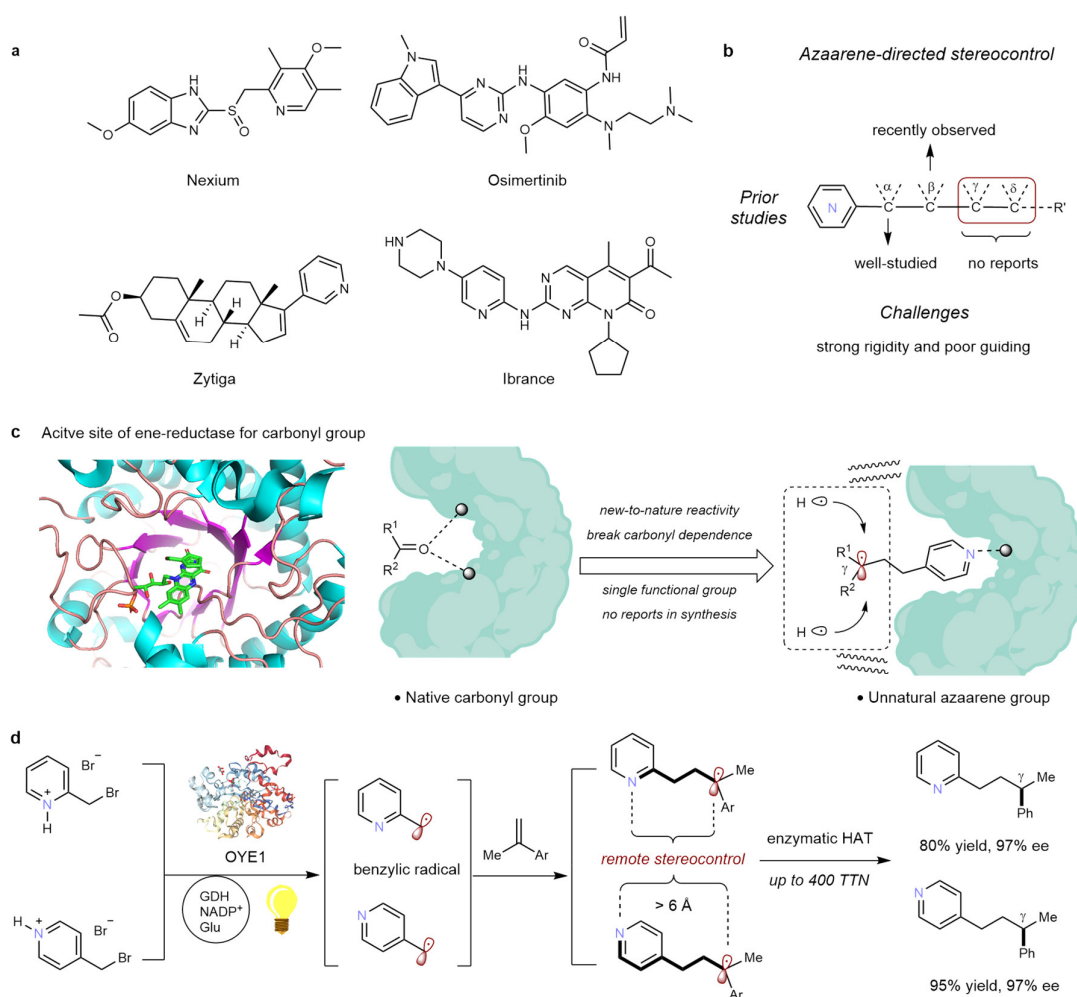


Fig. 1. | Challenges and strategies for remote stereocontrol with an azaarene. **a**, Representative drugs demonstrating the ubiquity of azaarenes in bioactive molecules. **b**, Prior studies and challenges of azaarene-directed stereocontrol. **c**, Enzymatic strategy for remote stereocontrol with an azaarene (Active site of OYE1 using PDB: 3TX9). **d**, Enantioselective radical hydroalkylation reaction demonstrating the remote stereocontrol with an azaarene. FG, functional group; Me, methyl; Ph, phenyl; Ar, aryl.

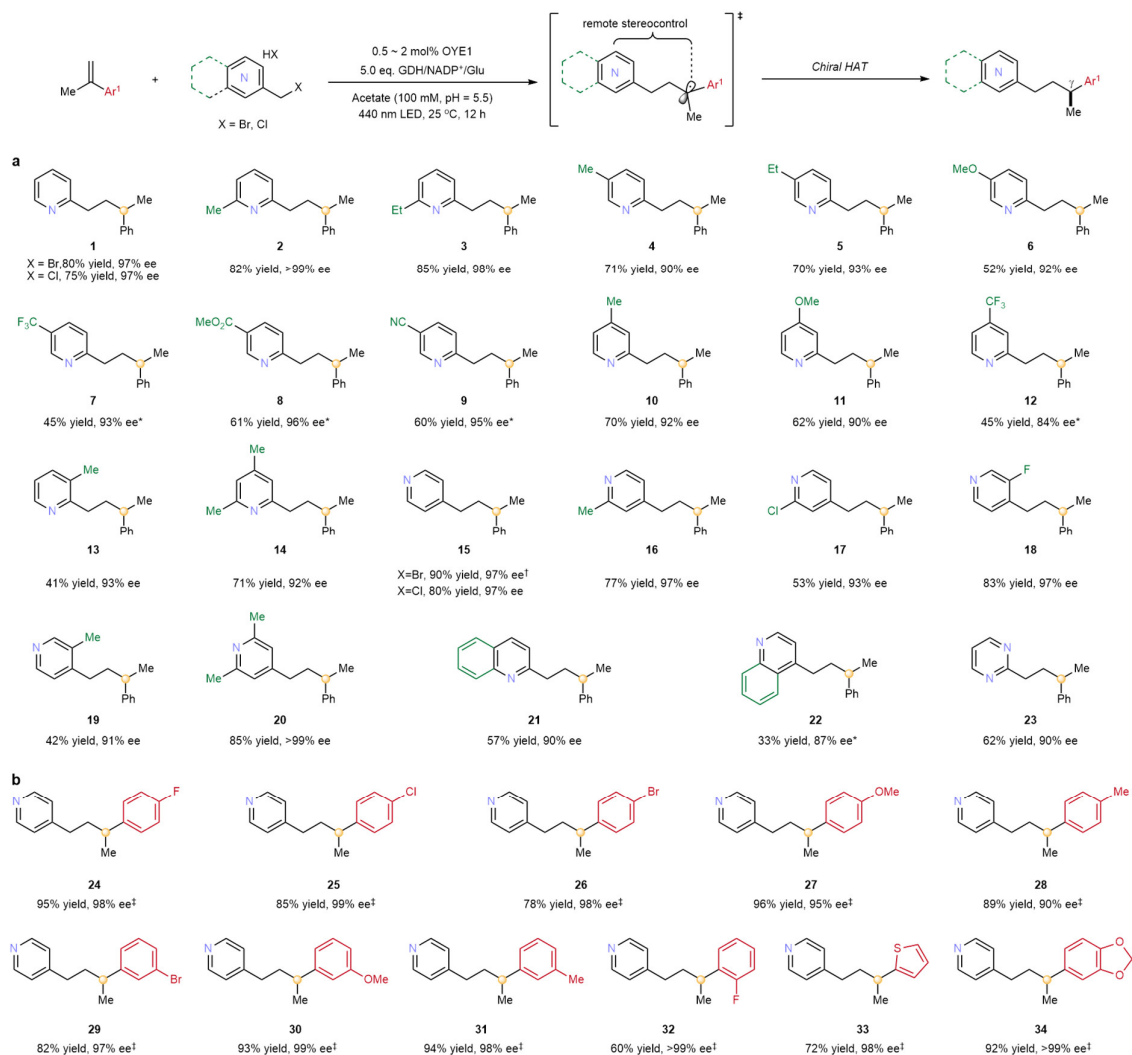


Fig. 2. | Scope of azaarenes and alkenes in the enantioselective radical hydroalkylation reaction. Reaction conditions: 2-bromomethyl pyridines (0.12 mmol) or 2-chloromethyl pyridines (0.12 mmol), α -methyl styrene (0.06 mmol), OYE1 (1 mol % based on α -methyl styrene), GDH-105 (2.0 mg), NADP⁺ (1 mol% based on glucose), glucose (0.3 mmol), 0.75 mL DMSO as cosolvent in 15 ml (total) 100 mM sodium acetate buffer pH = 5.5, 445 nm light-emitting diode (LED), 25 °C, 12 h. Isolated yields are given. The ee values were determined by high-performance liquid chromatography (HPLC). **a**, Scope of azaarenes. **b**, Scope of alkenes. *OYE1 (2 mol%). †OYE1 (0.5 mol%), ‡4-bromomethyl pyridines (0.06 mmol), alkenes (0.12 mmol). Me, methyl; Et, ethyl; OMe, methoxy; Ph, phenyl.

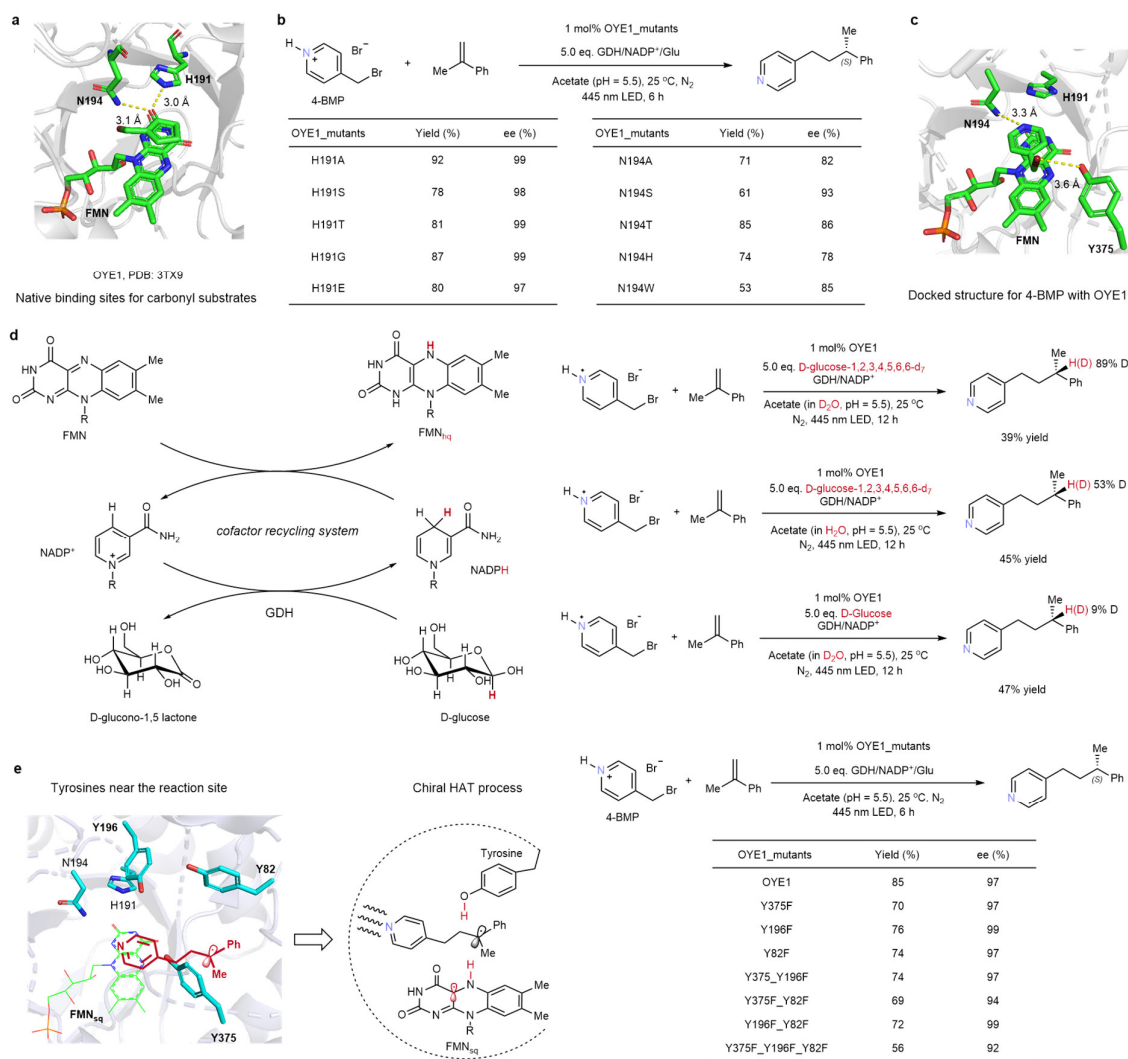


Fig. 3. | Detailed insights into mechanistic studies of OYE1. **a**, Illustration of the native binding sites H191 and N194 within OYE1 that are pertinent for carbonyl substrate attachment. **b**, Comprehensive assessment of the mutational impacts on the sites H191 and N194, indicating their roles in enzymatic activity. **c**, Visualization of potential interactions between the 4-BMP molecule and specific amino acid residues as deduced from molecular docking experiments. **d**, Depiction of the cofactor recycling mechanism alongside results from deuterium labeling experiments, elucidating the reaction pathway. **e**, Investigation into the crucial role played by tyrosine residues situated proximally to the reaction site, highlighting their influence on substrate turnover.

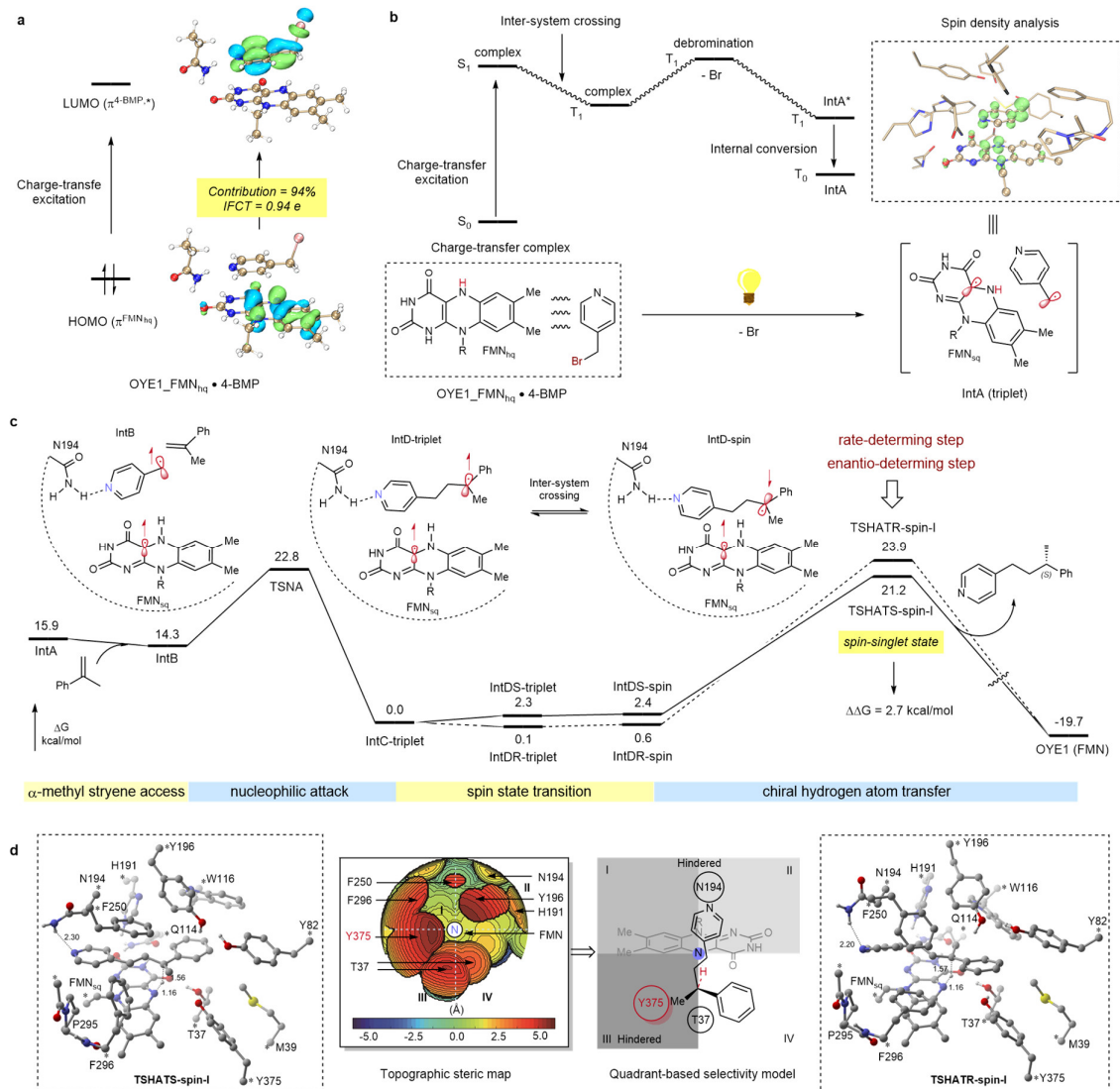


Fig. 4. | Insights from computational analyses of enzyme mechanisms. **a**, The illustration shows the dominant molecular orbital transitions contributing to charge-transfer excitation. Except for residue N194, other amino acid residues are omitted to enhance visual clarity. The color coding with green and blue represents positive and negative orbital phases, respectively. **b**, The diagram details the photochemical process that results in the formation of an aza-benzylic radical and FMN $_{sq}$. Spin density analysis of intermediate state **IntA** is also presented. Key states, including S₀ (ground state of singlet), S₁ (first excited state of singlet), T₀ (ground state of triplet), and T₁ (first excited state of triplet) are depicted and labeled. **c**, Calculated Gibbs energy profile for the nucleophilic attack step and HAT step. Calculations were performed at the um062x-D3/def2tzvpp (SMD, $\epsilon = 4.0$) // ub3lyp-D3(BJ)/def2svp (gas) level. Gibbs free energies of all structures are with respect to **IntC-triplet** (kcal/mol). Refer to Supplementary Fig. 5 for the structures of intermediates. **d**, Topographic steric maps for OYE1's catalytic pocket and DFT-optimized lowest-energy transition structures for *R* and *S* products. Note the atoms held constant during geometry optimization, indicated by an asterisk (*).

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Methods

Typical procedure for enantioselective radical hydroalkylation reaction (large scale). The acetate buffer (120 mM, 13 mL), GDH-105 (2.0 mg), NADP⁺ solution (2.2 mg, 0.003 mmol), glucose solution (54 mg, 0.3 mmol), 2-bromomethyl pyridine solution (15.2 mg, 0.06 mmol), and α -methyl styrene (14.2 mg, 0.12 mmol) were introduced into a 40 mL glass vial in an N₂-filled glovebox. After OYE1 (1 mol%) was added into the vial, the reaction mixture was subsequently loaded on the following photocatalytic setup and illuminated with 40 W 445 nm blue LED at 25 °C for 12 h. Afterwards, 5 mL saturated Na₂CO₃ were added to the reaction mixture, which was extracted with 2 x 10 mL EtOAc. The organic layer was dried over MgSO₄ and filtered to remove the drying agent. The residue was then purified by preparative TLC (ethyl acetate/hexane) to give the pure product with isolated yield.

Typical procedure for enantioselective radical hydroalkylation reaction (small scale). The acetate buffer (120 mM, 800 μ L), GDH-105 (~5U in 20 μ L), NADP⁺ solution (10 mM, 20 μ L), glucose solution (500 mM, 40 μ L), 2-bromomethyl pyridine solution (200 mM, 40 μ L), and α -methyl styrene DMSO solution (100 mM, 40 μ L) were introduced into a 4 mL glass vial in an N₂-filled glovebox. After OYE1 (1 mol%) was added into the vial, the mixture was stirred at 25 °C. The reaction mixture was subsequently loaded on the following photoreactor and illuminated with 540 mW 445 nm blue LED at

25 °C for 3 h. Afterwards, dodecane and 1 mL saturated Na₂CO₃ were added to the reaction mixture, which was extracted with 2 x 1 mL EtOAc and loaded on GC-MS for characterization. The GC yield was calculated according to the calibration curve constructed with pure racemates of the products.

Cloning, expression, purification and concentration determination of enzymes

Cloning and expression. The OYE1 ene-reductases employed in this research were originally isolated from baker's yeast. The plasmids **pET30a-OYE1** with His-tag were gifts from Dr. Francesco G. Gatti's lab. The plasmids were transformed into *Escherichia coli* BL21. For expression of OYE1, *E. coli* BL21 colony harboring **pET30a-OYE1** was inoculated in 5 mL LB containing 50 µg/mL kanamycin and grown overnight at 37 °C. These overnight cultures were used to inoculate 600 mL of TB medium containing 50 µg/mL kanamycin, which were grown at 37 °C until ODs of ~0.6 were reached. Protein expression was subsequently induced with the addition of 0.1 mM IPTG. The induced cultures were placed at 25 °C, 250 rpm, for 16 h for protein production.

Purification. Following expression, *E. coli* BL21 cells were collected by centrifugation (10,876 × g for 15 min) and resuspended with NaPi buffer, pH = 7.4. For the purifications of protein, FMN-Na (1 mg/mL) was added to the cell suspension. The mixture was then lysed by sonication and clarified by centrifugation (4 °C, 24,470 × g for 60 min). The protein was subsequently purified by affinity chromatography using two 5 mL HisTrap HP columns fitted to an ÄKTAexpress FPLC system (GE Health Life Sciences, Pittsburgh, PA). The purified proteins were buffer-exchanged against 100 mM NaPi buffer, pH 7.4 using an Amicon Ultra concentration tube with 30 kDa cut-off. The proteins were stored in 20% glycerol at -80 °C.

Concentration determination of enzymes. A 10 µL enzyme solution was added to an aqueous solution containing 980 µL water and 10 µL 10% wt% sodium dodecyl sulfate solution. The mixture was vortexed for 1 min and the absorbance at 446 nm ($\epsilon_{446\text{ nm}} = 12200\text{ M}^{-1}\text{cm}^{-1}$) was taken to calculate the concentration of the enzyme using the equation below:

$$c = \frac{Abs}{\epsilon l}$$

Site-directed mutagenesis for OYE1. Molecular biology reagents and chemicals were purchased from Fisher Scientific, Sigma-Aldrich, or New England Biolabs, Inc., if not specifically indicated. *E. coli* BL21 (DE3) cells (New England Biolabs, MA) were cultured in Luria-Bertani broth. DNA sequencing was performed at ACGT (Wheeling, IL). Primers were ordered from Integrated DNA Technologies (Coralville, IA). The plasmid **pET30a-OYE1** WT with C-terminal His-tag was used as the template for the polymerase chain reactions (PCRs) performed in the present work unless specified otherwise. Chemically competent *E. coli* BL21 (DE3) cells were used as hosts for plasmids amplification and protein expression, respectively. OYE1 variants were stepwise constructed by PCR according to the QuikChange site directed mutagenesis method⁴³. For detailed sequences of oligonucleotides, please refer to Supplementary Methods.

Cluster approach Setup. The initial structure was derived from the crystal structure (PDB code: 3tx9), which includes 2-(Hydroxymethyl)-cyclopent-2-enone. A cluster model was established by extracting residues Thr37, Met39, Tyr82, Gln114, Trp116, His191, Asn194, Tyr196, Arg243, Phe250, Pro295, Phe296, Tyr375, and the flavin mononucleotide (FMN) from the crystal structure. The phosphate portion of FMN was excluded and terminal atoms were fixed. In terms of amino acid residues, terminal carbons were fixed during geometry optimizations to restrict unrealistic residue movements. Subsequently, 2-(Hydroxymethyl)-cyclopent-2-enone was eliminated from the system and its binding

position was substituted with 4-bromomethyl pyridine. AutoDock Vina was employed to dock 4-bromomethyl pyridine into the active site post ligand removal. We considered three possible bindings of 4-bromomethyl pyridine (to H191, N194, or both H191 and N194), with a total of six conformations (Supplementary Fig. 3). The deprotonation states of H191 were assigned based on pK_a values obtained from the PROPKA⁴⁴, complemented with meticulous visual inspection of local hydrogen-bonded networks. When α -methyl styrene was involved in subsequent reactions, the dissociated bromide was removed from the system to generate carbon radical intermediates (**IntA**) with six total conformations (Supplementary Fig. 4). For detailed calculation methods, please refer to Supplementary Methods.

DFT method. DFT calculations were carried out using the Gaussian 16 program package, Revision C.02 software⁴⁵. The B3LYP method^{46,47} and the dispersion energy corrections by D3 version of Grimme's dispersion with the original D3BJ damping function⁴⁸ were used. The def2svp basis set^{49,50} of Ahlrichs and coworkers for all the atoms has been selected for geometry optimizations and calculation of Gibbs energy corrections at 298 K temperature in the gas phase. Single-point energies were calculated at the M062x-D3 level⁵¹ of theory with def2tzvpp for all atoms with the value of the dielectric constant $\epsilon = 4$. The SMD solvation model⁵² was used to model the effects of the rest of the enzyme that was not included in the model. Energies reported herein are thus those for the large basis set (which include dispersion) corrected for zero-point energies (ZPE) and solvation effects. Reaction paths were traced by the intrinsic reaction coordinate method for all transition states.

TDDFT method. The initial structures of CT complex were first optimized at B3LYP-D3BJ/def2svp level. Then, the single-point TDDFT calculations at wB97XD/def2-TZVP level⁵³ were performed to compute the five low-lying singlet excited states. All TDDFT calculations were performed with Gaussian 16, Revision C.02. The software Multiwfn 3.8 was used for wave function analysis.

Topographic steric maps generation. Topographic steric maps were generated using the SambVca online tool (<https://www.molnac.unisa.it/OMtools/sambvca2.1/index.html>). The process began with the single crystal structure of the protein (PDB ID: 3tx9). After removing the ligands, amino acid residues within a 7 angstrom radius of the FMN were selected for input into the SambVca tool for steric topographic mapping. To set up the map, hydrogen atom was added to the nitrogen atom of the FMN. The N-N-H angle of the FMN was adjusted to 127 degrees to align with the angle of chiral hydrogen atom transfer as determined from DFT calculations. This setup established the N-H direction as the Z-axis and the FMN plane as the XZ plane for the steric map. To accurately represent the steric effect within the active site, residues W116 and Y82 were removed from the model. Although these residues are in proximity to FMN, they are positioned above it and do not contribute to steric hindrance when the substrate is in close contact with FMN to acquire a hydrogen atom. The exclusion of these residues ensures that the map more accurately reflects the steric environment encountered by the substrate.

Data availability. Data relating to the materials and methods, optimization studies, experimental procedures, calculations, atomic coordinates, HPLC spectra and NMR spectra are available in the Supplementary Information. The structure of OYE1 used for our calculations is available in the PDB database (3TX9). All data is available from the authors upon reasonable request.

Methods-only references

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